Methods in Pseudo-Boolean Optimization

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Outline

1. Formulating pseudo-Boolean functions
2. Application to lower bounding quantum query complexity
3. Comparing approximate optimization algorithms

Definition

Definition (Pseudo-Boolean function)

$$f : \{0,1\}^n \to \mathbb{R}$$

Example (Boolean function)

$$f : \{0,1\}^n \to \{0,1\}$$

Definition (Pseudo-Boolean optimization)

$$x^* := \arg\min_{x \in \{0,1\}^n} f(x)$$

Polynomial Unconstrained Boolean Optimization (PUBO)

Theorem

Every pseudo-Boolean function f can be represented uniquely as a multilinear polynomial

$$f(x) = \sum_{S \in \mathcal{N}} c_S \prod_{i \in S} x_i$$

where $\prod_{i \in \emptyset} x_i := 1$ *and* $\mathcal{N} := \mathcal{P}(\{1, \ldots, n\})$

- Multilinear because $x_i^2 = x_i$
- $\deg f = \max_{S \in \mathcal{N} : c_S \neq 0} |S|$ (unless f is zero, then $\deg f := -\infty$)

From PUBO to Hamiltonian

- The group $(\{0,1\},\oplus)$ is isomorphic to $(\{1,-1\},\times)$ via the map $\phi:\{0,1\}\to\{1,-1\}$, $x\mapsto 1-2x$, $\phi^{-1}:s\mapsto(1-s)/2$
- Let $f(x)=f(x_1,\ldots,x_n)$ be a PUBO
- Define $H:\{1,-1\}^n\to\mathbb{R}$ by $H(\sigma^z)=f\left(\phi^{-1}\left(\sigma_1^z\right),\ldots,\phi^{-1}\left(\sigma_n^z\right)\right)$
- Promote H to a Hamiltonian $\hat{H}$ by promoting spins to Pauli-Z operators $\hat{\sigma}_i^z$
- Then $\hat{H}\left|x\right\rangle=f(x)\left|x\right\rangle$, where $\{\left|x\right\rangle\}$ is the computational basis $(\left|\uparrow_z,\downarrow_z\right\rangle^{\otimes n})$

$$\deg f=k \;\leftrightarrow\; \hat{H} \text{ is } k\text{-local}$$

Unique PUBO corresponding to pseudo-Boolean function

$f(x) = \sum_{S \in \mathcal{N}} c_S \prod_{i \in S} x_i$

- Clearly the $c_\emptyset$ coefficient is uniquely determined as $c_\emptyset = f(0^n)$
- To set the remainder of the coefficients, consider defining f by its "truth" table

Example ($n = 2$)

$$f(x) = (1 - x_1)(1 - x_2)f(0,0) + (1 - x_1)x_2 f(0,1) + x_1(1 - x_2)f(1,0) + x_1 x_2 f(1,1)$$

Some Boolean formulas

- $\mathrm{OR}(x_1, x_2) = (1 - x_1)(1 - x_2)(0) + (1 - x_1)x_2(1) + x_1(1 - x_2)(1) + x_1x_2(1) = x_1 + x_2 - x_1x_2$
- $\mathrm{OR}(x_1, x_2, x_3) = \mathrm{OR}(x_1, \mathrm{OR}(x_2, x_3))$
- $\mathrm{AND}(\mathrm{x}) = \prod_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{x_i}$
- $\mathrm{XOR}(\mathrm{x}_1, \mathrm{x}_2) = (\mathrm{x}_1 - \mathrm{x}_2)^2$
- $\mathrm{NOT}(\mathrm{x}_1) = 1 - \mathrm{x}_1$
- $\mathrm{NOR}(x) = \mathrm{NOT}(\mathrm{OR}(x))$

PUBO is NP-hard

Example (Reduce 3-SAT to PUBO)

$$\begin{aligned} \text{3-SAT}: \quad & x \mapsto (x_1 \vee \neg x_2 \vee \neg x_3) \wedge (\neg x_1 \vee \neg x_2 \vee x_4) \wedge \dots \\ f: \quad & x \mapsto 1 - \mathrm{AND}\left[\mathrm{OR}\left(x_1, \mathrm{NOT}(x_2), \mathrm{NOT}(x_3)\right), \mathrm{OR}\left(\mathrm{NOT}(x_1), \mathrm{NOT}(x_2), x_4\right), \dots\right] \end{aligned}$$

Satisfying $x \leftrightarrow f(x) = \min f = 0$

Notice f is a Boolean function

Degree ≥ 3 PUBO is NP-hard

- The previous encoding of 3-SAT will in general have $\deg f \sim n$
- We'd like to think about k-local Hamiltonians, so we want $\deg f$ to be a constant
- Adding constraints to a PUBO corresponds to adding penalty terms

Example (Reduce 3-SAT to degree 3 PUBO)

$$\begin{aligned} \text{3-SAT}: \quad & x \mapsto (x_1 \vee \neg x_2 \vee \neg x_3) \wedge (\neg x_1 \vee \neg x_2 \vee x_4) \wedge \dots \\ f: \quad & x \mapsto \mathrm{NOR}\left(x_1, \mathrm{NOT}(x_2), \mathrm{NOT}(x_3)\right) + \mathrm{NOR}\left(\mathrm{NOT}(x_1), \mathrm{NOT}(x_2), x_4\right) + \dots \end{aligned}$$

Satisfying $x \leftrightarrow f(x) = \min f = 0$

Notice f is *not* a Boolean function

QUBO is NP-hard

- Quadratic Unconstrained Boolean Optimization (QUBO) problems are degree 2 PUBOs (i.e. 2-local, think Ising model)
- Can we reduce the degree 3 PUBO representing 3-SAT to a QUBO?

Theorem

If k is a constant, then any degree k PUBO can be reduced to QUBO with poly(n) overhead

Therefore, 3-SAT can be reduced to QUBO $\rightarrow$ QUBO is NP-hard

Degree reduction

- Let g be an arbitrary QUBO
- Consider $f(x) = g(x) + cx_1x_2x_3$
- Let $h(a, b, c) = 3a - 2a(b + c) + bc$, note h is a QUBO and

$$h(a, b, c) \begin{cases} = 0 & a = bc \\ > 0 & \text{else} \end{cases}$$

- There exists a sufficiently large λ such that

$$\min_{x \in \{0,1\}^n} f(x) = \min_{x \in \{0,1\}^n} \min_{y \in \{0,1\}} g(x) + cx_1y + \lambda h(y, x_2, x_3)$$

Optimal degree reduction

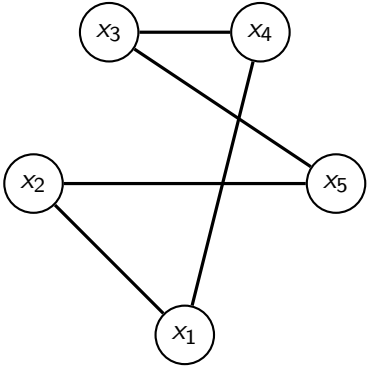
- Use the same ancilla variable to reduce multiple terms
- Find pairs of x_i variables that are together most frequently

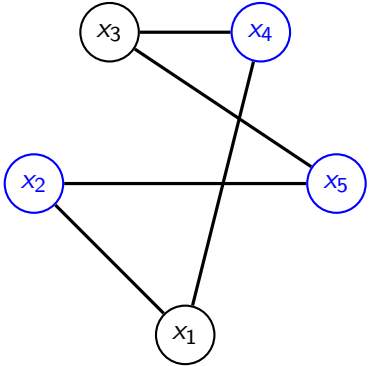
Theorem

Finding the optimal set of pairs that minimizes the number of ancillas needed to reduce the degree of PUBO is NP-hard

Optimal degree reduction is NP-hard

$f(x) = x_0\,(x_1x_2 + x_1x_4 + x_2x_5 + x_3x_4 + x_3x_5)$





Vertex Cover (NP-hard)

$f(x) = x_0\,(x_1x_2 + x_1x_4 + x_2x_5 + x_3x_4 + x_3x_5)$

Optimal pairs are x_0x_i

Encoding constraints

Example

$$\text{Profit} = \max_{x \in \{0,1\}^{24 \cdot 7}} \sum_{i=1}^{24 \cdot 7} p_i x_i \quad \text{subject to the constraints:}$$

1. $\sum_{i=1}^{24 \cdot 7} x_i = 40$ (40 hour work week)
2. $\forall d \in \{0, \ldots, 6\} : \quad \sum_{i=24d}^{24(d+1)} x_i \leq 10$ (ten hour workday or less)

Represent as PUBO $\leftrightarrow$ encode constraints

Equality constraints

- $\min_x f(x)$ subject to $\sum_i x_i = H$

$$\min_x f(x) + \lambda \left(\sum_i x_i - H \right)^2$$

- Taking λ large strictly enforces the constraint
- But also creates a jagged optimization landscape

Inequality constraints

#	Penalty	Coefficient range	Ancilla
1	$\left(\sum_{t=0}^{N-1} x_t + \sum_{j=0}^{m} 2^j y_j - H\right)^2$	$[-H2^{m+1}, 2^{2m+1}]$ $\sim [-H^2, H^2]$	$m+1$
2	$\left(\sum_{t=0}^{N-1} x_t + \sum_{j=0}^{H-1} y_j - H\right)^2$	$[-2H, 2]$	H
3	$\left(\sum_{t=0}^{N-1} x_t - \sum_{j=0}^{m-1} 2^j y_j - \sum_{j=m}^{H-2^{m-1}+m-1} y_j\right)^2$	$[-2^m, 2^{2m-2}]$ $\sim [-H, H^2]$	$m + H - 2^{m-1}$
4	$\left(\sum_{t=0}^{N-1} x_t - \sum_{j=0}^{H-1} y_j\right)^2$	$[-2,2]$	H
5	$\left(\sum_{t=0}^{N-1} x_t - y_0 - 2y_1 - 4\sum_{j=2}^{1+\lfloor H/4 \rfloor} y_j\right)^2$	$[-8,16]$	$2 + \left\lfloor \frac{H}{4} \right\rfloor$

$$\sum_i x_i \leq H$$

$$m = \lceil \log_2(H+1) \rceil$$

Possible PUBO formulation

$$\text{Profit} = \max_{x \in \{0,1\}^{24\cdot 7}} \sum_{i=1}^{24\cdot 7} p_i x_i \quad \text{subject to the constraints:}$$

1. $\sum_{i=1}^{24\cdot 7} x_i = 40$ (40 hour work week)
2. $\forall d \in \{0, \ldots, 6\}: \quad \sum_{i=24d}^{24(d+1)} x_i \leq 10$ (ten hour workday or less)

$$\text{Profit} = -\min_{x \in \{0,1\}^{24\cdot 7}} \min_{y^{(0)} \in \{0,1\}^{10}} \cdots \min_{y^{(6)} \in \{0,1\}^{10}} \sum_{i=1}^{24\cdot 7} (-p_i) x_i$$

$$+ \lambda \left(\sum_{i=1}^{24\cdot 7} x_i - 40 \right)^2 + \sum_{d=0}^{6} \lambda^{(d)} \left(\sum_{i=24d}^{24(d+1)} x_i - \sum_{j=1}^{10} y_j^{(d)} \right)^2$$

NP-hard vs NP-complete

Which one is harder?

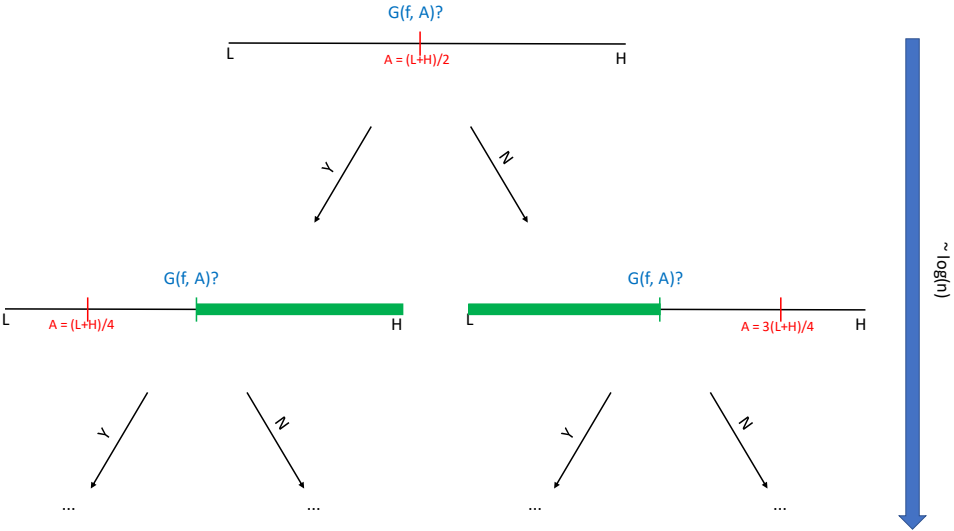
Given a degree $k \geq 2$ pseudo-Boolean function f:

- NP-hard: minimize f
- NP-complete: is $\min f < A$?

- $\min f \geq \sum_{S \in \mathcal{N}} c_S \Theta(-c_S) \equiv L$
- $\max f \leq \sum_{S \in \mathcal{N}} c_S \Theta(c_S) \equiv H$
- $\max f - \min f \leq H - L \sim \mathcal{O}(n^k)$ assuming $c_S \sim \mathcal{O}(1)$

Let G be an oracle that solves the NP-complete problem; $G(f, A) = (\text{is } \min f < A)$

NP-hard with an NP-complete oracle



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Approximate degree of a Boolean function

Boolean function $f : \{0,1\}^N \to \{0,1\}$ e.g. $f(x_1, \ldots, x_N) = \mathrm{OR}(x_1, \ldots, x_N)$

Definition (ϵ-approximation)

G_ϵ is the set of all **pseudo-Boolean** functions $g : \{0,1\}^N \to [0, \epsilon) \cup (1-\epsilon, 1]$ that satisfy

$$\forall x \in \{0,1\}^N : |f(x) - g(x)| < \epsilon$$

Definition (approximate degree)

$$\widetilde{\deg}_\epsilon f := \min_{g \in G_\epsilon} \deg g$$

Quantum query complexity and Boolean functions

We query the quantum oracle $i \mapsto x_i^*$ a total of T times

Theorem

To determine $f(x^*)$ with probability 1, $T \in \Omega(\deg f)$ e.g. $\deg \mathrm{OR} = N$

Theorem

To determine $f(x^*)$ with ϵ-bounded error, $T \in \Omega\left(\widetilde{\deg}_\epsilon f\right)$

Chernoff's bound $\rightarrow$ w.l.o.g. $\epsilon < 1/2$

Theorem (Aaronson et al. 23 Oct. 2020 (arXiv:2010.12629))

For any Boolean function f, $\deg f \in \mathcal{O}(\widetilde{\deg}(f)^2)$ e.g. Grover $\in \Theta\left(\sqrt{\deg \mathrm{OR}} = \sqrt{N}\right)$

https://www.cs.umd.edu/~amchilds/qa/qa.pdf

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Comparing approximate optimization algorithms

- $f : \{0,1\}^n \to \mathbb{R}$
- $\mathrm{QAOA}_p(H_f)$ is depth-p QAOA
- $\rho_p(x) = \left|\langle x| \mathrm{QAOA}_p(H_f) |+\rangle^n\right|^2$

Is ρ_{p+1} better than ρ_p?

Yes? $\langle f \rangle_{\rho_{p+1}} \leq \langle f \rangle_{\rho_p}$

But we may be interested in higher moments $\langle f^m \rangle$

First thing to check: is my algorithm better than random sampling?

Relaxing the domain

- $f : \{0,1\}^n \to \mathbb{R}$
- Then $f^2 : \{0,1\}^n \to \mathbb{R}$ satisfies $f^2(x) = f(x)^2$
- Consider relaxing the domain of f, is $f^2(x) = f(x)^2$ for all $x \in [0,1]^n$?

No! (recall multilinearity)

$$f(x \in \{0,1\}) = x \qquad f(x \in [0,1]^n)^2 = x^2$$
$$f^2(x \in \{0,1\}) = x^2 = x \qquad f^2(x \in [0,1]^n) = x$$

Expectation values of PUBOs

$\langle f^m \rangle_{\text{uni}} = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} f(x)^m$ *time* $\sim 2^n$

Theorem

$\langle f^m \rangle_{\text{uni}} = f^m \left(\frac{1}{2}, \frac{1}{2}, \dots\right)$ *time* $\sim n^{km}$

Example (Variance)

$\mathrm{Var}_{\text{uni}}(f) = f^2 \left(\frac{1}{2}, \frac{1}{2}, \dots\right) - f \left(\frac{1}{2}, \frac{1}{2}, \dots\right)^2$

Proof.

$$\langle f^m \rangle_{\text{uni}} = \left\langle \sum_{S \in \mathcal{N}} c_S \prod_{i \in S} x_i \right\rangle_{\text{uni}} = \sum_{S \in \mathcal{N}} c_S \left\langle \prod_{i \in S} x_i \right\rangle_{\text{uni}} = \sum_{S \in \mathcal{N}} c_S \prod_{i \in S} \langle x_i \rangle_{\text{uni}}$$

□

Expectation value of random PUBOs

- What if every coefficient is drawn from the same distribution σ?
- $\langle f^m \rangle_{\mathrm{uni}} = \langle c \rangle_\sigma^m \langle F_{n,k}^m \rangle_{\mathrm{uni}}$

Example

$$F_{n,2}(x) = \sum_{i=1}^{n} x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} x_i x_j$$

What is $\langle F_{n,k}^m \rangle_{\mathrm{uni}}$?

Symmetric case

- Redefine (just for this slide) a pseudo-Boolean function to be symmetric

Example

$$F_{n,2}(x) = \sum_{i=1}^{n} x_i + \sum_{i=1}^{n} \sum_{j=1 \neq i}^{n} x_i x_j$$

- $F_{n,k}(x) = \left(\sum_{i=1}^{n} x_i\right)^k$
- $S(r,t)$ – Stirling's numbers of the second kind – number of ways to distribute r *identical* objects into t *identical* bins *time* $\sim t$

Theorem

$$\left\langle F_{n,k}^m \right\rangle_{\text{uni}} = F_{n,k}^m\left(1/2, 1/2, \dots\right) = \sum_{i=1}^{km} \frac{1}{2^i} \frac{n!}{(n-i)!} S(km, i) \qquad \textit{time} \sim (km)^2$$

References

- Endre Boros and Peter L. Hammer. *Pseudo-Boolean optimization*. Discrete Applied Mathematics 123, 1, 155-225 (2002).